



Jonish Abisheck Joseph

B.E Mechanical

CGPA : 8.68

Birla Institute of Technology and Science, Pilani – Goa Campus

LinkedIn

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EDUCATION

- Under Graduate: Birla Institute of Technology and Science, Pilani – Goa Campus** 2021 - 2025
B.E Mechanical, Goa CGPA: 8.68/10
- Minor in Robotics and Automation: BITS Goa** Aug 2023 - May 2024
Courses: Robotics, AI for Robotics, Control Systems, Mechanisms and Machines, Machine Learning CGPA: 9.2/10
- Online BSc Data Science Course (Foundational Level): IIT Madras** Jun 2021 - Aug 2022
Courses: Probability & statistics, Python, english CGPA: 8.6/10

AREAS OF INTEREST

- Autonomous Navigation and Control
- Robot Kinematics & Locomotion
- Path Planning Algorithms
- Reinforcement Learning

EXPERIENCE

- Comprehensive testing and validation of a custom path planning algorithm.** Jul 2025 - present
IIT Delhi, Research Assistant - Guide: Prof. Dr. Subir Kumar saha Delhi, India
 - Worked on a Differential Drive Non-Holonomic Mobile Robot built on ROS.
 - Tested the Robot on various Maze setups and tuned the TEB Path planner and Extended Kalman Filter Parameter for smooth Navigation.
- Modeling and control of flapping wing Unmanned Aerial Vehicles (UAVs).** Jun 2024 - Dec 2024
IIIT Hyderabad, Research Intern - Guide: Prof. Dr. Harikumar Kandath Hyderabad, India
 - Worked on a 3 DOF Flapping wing MAV, Which is RC controlled using Ardu Pilot.
 - Developed a novel mathematical model for flapping wing MAVs that integrates aerodynamic principles from both fixed-wing and flapping-wing flight regimes.

PROJECTS

- Dynamic Force Analysis of a transtibial prosthetic user through 2R serial Manipulator.** May 2023 - May 2024
BITS Goa, Research Project – Guide: Prof. Dr. Ganesh Bapat
 - * Developed a 2R serial robot model in MATLAB and conducted force analysis to determine the force acting on the Stump of the Patient while walking.
 - * Performed Finite Element Analysis (FEA) in Ansys to predict socket deformation under various applied forces.
- Benchmark testing of the Pipe Network (PINET) code for nuclear reactor analysis.** Jun 2023 - Jul 2023
Indira Gandhi Centre for Atomic Research[IGCAR], Research Project - Guide Mr. Vikram Govindarajan
 - * Developed a Python-based Framework on PINET to simulate the reactor core, applying heat transfer principles for accurate analysis.
 - * Conducted six benchmark tests using the developed model, with results matching with 90% accuracy.

TECHNICAL SKILLS

Programming Languages: C++, Python.

Frameworks: ROS, Matlab, Tensorflow.

Design and Simulation: Gazebo, Solidworks, Ansys.

Relevant Coursework: Robotics, AI for Robotics, Control Systems, Machine Learning.

PAPER PUBLICATION

- Simulation of Operational Transients Performed During Startup Tests of Superphénix Reactor Using PINET Code [LINK]** ISHMT-ASTFE 2024 - Accepted
Authors: J Iniya Adhitya, J Jonish Abisheck, G Vikram, AK Chauhan

ACHIEVEMENTS

- GATE 2025 - All India Rank 170** Feb 2025
Secured an All India Rank of 170 in the GATE 2025 Examination for Mechanical Engineering.